

Evaluation Problem 2

Group C22

Duration: 2 hrs

Instruction:

Do not use help from internet. Your code will be checked for plagiarism. Do not share your code with friends, in case of plagiarism both parties will get 0.

Implement a Preemptive Priority Scheduler with Aging. This prevents "Starvation" where low-priority processes never run.

- The scheduler always picks the process with the highest priority available at the current time.
- If a new process arrives with a higher priority than the currently running one, the current process is preempted.
- Every 5 units of time a process spends in the ready queue (waiting), its priority should be increased by 1 (e.g., priority 5 becomes 4).
- If two processes have the same priority, then select the process who was ready first (FCFS).

Input: A list of processes with Unique process number(integer), Arrival Time, Burst Time, and Initial Priority (lower number = higher priority).

Output:

A Gantt Chart representation (text-based).

Average Waiting Time (AWT).

Average Turnaround Time (ATAT).

Marks distribution

1. Gantt chart for Priority scheduling with pre-emption. **(10 marks)**

*Without pre-emption 5 marks

2. Implementation aging. **(10 marks)**

3. Average wait time and turnaround time. **(10 marks)**